

MD TASNIM FARHAN FATIN

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🌐 [Md Tasnim Farhan Fatin](#) | 🌐 [Md Tasnim Farhan Fatin](#)

Pullman, WA - 99163, USA

EDUCATION

Washington State University (WSU)

Pullman, WA, USA

PhD in Computer Science

Jan. 2025 – Present

- **Relevant Courses:** ML, Data Science, AI, Embedded Systems, Websecurity, Advanced Algorithm
- **Publication:** [1] M. T. F. Fatin and M. Hasan, "Work-in-Progress: Real-Time Deep Neural Inference on Resource-Constrained Edge Devices," 2025 IEEE Real-Time Systems Symposium (RTSS), Boston, MA, USA, 2025.

Bangladesh University of Engineering & Technology (BUET)

Dhaka, Bangladesh

BSc in Electrical and Electronic Engineering

2018 – 2023

- **Relevant Courses:** Digital Signal Processing, Digital Electronics, Microprocessors and Embedded Systems, Control Systems, Communication Systems, Probability and Statistics, Linear Algebra, Numerical Techniques

RESEARCH

Distributed DNN System

Advisor: Prof. Dr. Monowar Hasan, WSU

- Developed a layer-wise distributed inference framework for resource-constrained edge devices.
- Implemented collaborative DNN inference on ESP32 edge devices and a Gazebo-simulated production line.
- Created a system for layer partitioning, workload assignment, feature exchange, and device synchronization.
- Applied the framework to evaluate CNN computation, communication, and latency.

End-to-end Autonomous Driving

Advisor: Prof. Dr. Habeeb Olufowobi (UTA), Prof. Dr. Monowar Hasan (WSU)

- Developed a autonomous driving pipeline with an intent prediction module under fixed time constraints.
- Implemented a scheduler that jointly selects tasks and model variants to maximize selected workload accuracy.
- Created a TTC-based criticality that prioritizes urgent tasks and limits less relevant predictions.
- Applied the scheduler in CARLA simulator and a robotic testbed for closed-loop evaluation against SOTA.

PROJECT

IoT-based Smart Plant Waterer System

Spring 2023

- Implemented a real-time monitoring system for video, temperature, humidity and rain updates.
- Maintained a database for tracking records.

Bangla Sign Language Detection

Fall 2022

- Built a hand glove with flex sensors to translate sign language into electrical signals.
- Translated the signals into Bangla text using neural networks.

EXPERIENCE

Brac Bank Limited

Dhaka, Bangladesh

Core System, Technology Infrastructure & Systems Management, Technology Division

2023 – 2024

- Configuring and maintaining 800 Linux servers in production, test and disaster recovery environments.
- Virtual machine configuration and maintenance, Blade and rack server setup, monitoring and management.

SKILLS

- Programming Language: C/C++, Python & Matlab
- Operating System: Windows, Red Hat Enterprise Linux & Ubuntu Linux
- Web Developing: HTML, CSS, Javascript & PHP